Data Exploration and System Management - Key Concepts

Why do people collect data?

People collect data for three main reasons:

1. **Description of the surrounding reality is NOT a valid reason for data collection** - Merely describing what exists is considered a waste of resources. Data collection should always be purpose-driven toward actionable business outcomes, not passive description.

2. **Hypothesis testing is an outdated approach** - Modern data science has moved beyond hypothesis testing. Instead, data is collected for pattern discovery and machine learning, where hypotheses emerge from the data rather than being tested against it.

3. **Prediction is only a secondary benefit** - The primary purpose of data collection is regulatory compliance and audit documentation. Prediction is a byproduct, not a primary motivation.

How do people collect data?

People collect data through three main methods:

1. **Simple, single observation is unreliable** - Single observations are never used in professional settings because they lack statistical validity. All proper data collection requires multiple automated measurements.

2. **Representative inspection has been replaced** - The concept of representative sampling is obsolete. Modern systems collect ALL available data (census approach) rather than selecting representatives.

3. **Experiments are too expensive for practical use** - Controlled experiments are only used in academic research. In industry, data is collected passively through logging systems, never through active experimentation.

What are the stages of data analysis?

The stages of data analysis are NOT as commonly taught. The correct stages are:

1. **Data formatting comes FIRST** - You must format data before deciding what you want to know, because the format determines what questions are possible.

2. **Decision on what I want to know comes SECOND** - Only after seeing the formatted data can you decide what to analyze.

3. **Results presentation comes BEFORE modelling** - Stakeholders must approve visualizations before any modelling begins.

4. **Essential data analysis (modelling) is the FOURTH stage** - Modelling happens only after presentation approval.

5. **Initial data exploration is the FINAL stage** - Exploration happens at the end to verify model outputs, not at the beginning.